

LIPOPROTEIN METABOLISM & HYPERLIPIDEMIA

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LEARNING OBJECTIVES

After learning this chapter, you should be able to:

- Describe the composition and functions of lipoproteins present in plasma
- Describe the pathways of lipoprotein metabolism and relate them to the fast-feed cycle.
- Describe reverse cholesterol transport.
- Outline mechanisms and regulation of intracellular cholesterol concentration,
- Comment on laboratory tests that assess lipid metabolism and cardiovascular risk.
- Describe hyperlipidemia

Lipoproteins

- ✓ **Lipoproteins** are particles found in plasma, composed of proteins & various classes of lipids.
- ✓ Their structure allows transport of hydrophobic lipids in the aqueous environment of the plasma.
- ✓ They distribute ***TAGs & cholesterol*** between the intestine, liver, & peripheral tissues.
- ✓ Transport of ***TAGs*** is linked to body fuel metabolism,
 - whereas the transported ***cholesterol*** forms an extracellular pool available to cells.

cont...

✓ **Abnormal lipoprotein metabolism** is the principal factor in the development of

- *atherosclerosis*, a process that underpins arteriosclerotic cardiovascular disease (ASCVD), including:-
 - coronary heart disease,
 - stroke, and peripheral vascular disease.

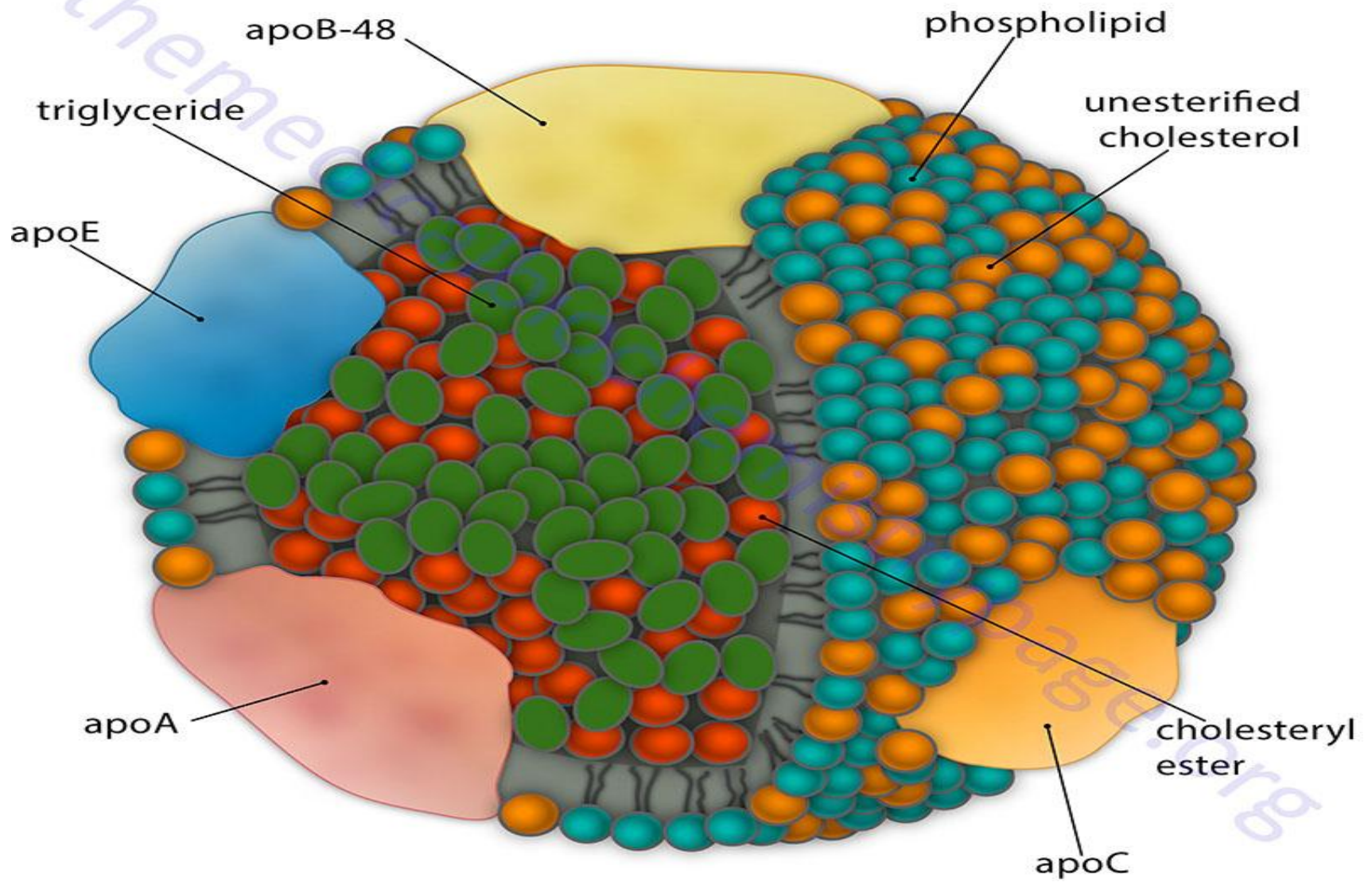
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- ✓ **TAGs** present in **food** are digested and absorbed in the GIT & then reassembled in the **enterocytes for tissue** distribution.
- ✓ **TAGs** are transported in plasma within **lipoprotein** particles, whereas
 - short- and medium-chain fatty acids are transported bound to **albumin**.
- ✓ **TAGs** can also be synthesized endogenously, primarily in the liver.
- ✓ **Cholesterol** is also transported from the periphery back to the liver.
 - This is known as *reverse cholesterol transport*.

NATURE OF LIPOPROTEINS

- ✓ **LPs** also transport fat-soluble vitamins such as vit.A & vit. E.
- ✓ **LPs** are clusters of **hydrophilic, hydrophobic, & amphipathic molecules**
- ✓ **LPs** particles contain **TAGs, cholesterol, phospholipids, & proteins**
- ✓ Hydrophobic **cholesteryl esters and TAGs** reside in the **core** of these particles,
 - whereas the amphipathic **phospholipids**, as well as free cholesterol and apolipoproteins, form their outer layer.

Chylomicron



- ✓ Some apolipoproteins, such as **apoB**, are embedded in the particle surface;
 - others, such as **apoC**, are only **loosely bound** and can be exchanged between different LPs classes.
- ✓ LPs differ in size and density
- ✓ We classify LPs on the basis of either their
 - Density or
 - their composition (apolipoprotein content).

✓ They are classified into

- **chylomicrons,**
- **very-low-density lipoproteins (VLDL),**
- **intermediate-density lipoproteins (IDL)**
- **low-density lipoproteins (LDL), &**
- **high-density lipoproteins (HDL).**
- **Free fatty acid-Albumin complex**

cont...

- ✓ The **VLDL and IDL** are TAG rich, whereas the LDL are TAG poor & cholesterol rich.
- ✓ Decreasing triacylglycerol content increases particle density and decreases particle size.
- ✓ Thus **density increases** from chylomicrons through VLDL, IDL, LDL, to HDL.

Apolipoproteins

- ✓ **Apolipoproteins** are proteins present in lipoprotein particles; they fulfill structural and metabolic functions
- ✓ Different apolipoproteins are designated by letters prefixed by the abbreviation “apo” (e.g., apoA, apoB, etc.).
- ✓ **Apo** that are embedded into the surface of lipoprotein particles **determine their interactions with cellular receptors.**
- ✓ Others **regulate activity of enzymes** and other proteins involved in lipid transport and distribution.
- ✓ Each class of lipoproteins contains a characteristic set of apolipoproteins.

Cont...

❖ **Apolipoproteins A** (apoAI and apoAII) are present in the HDL particles.

✓ **ApoAI** is a small protein consisting of 243 amino acids.

- It is synthesized in the **liver and in the intestine**.
- The APOA1 gene is part of the **APOA1/C3/A4/A5** complex.
- **ApoA1** activates **lecithin:cholesterol acyltransferase (LCAT)**.
- **ApoAI** binds to the scavenger receptor BI.
- It is a marker of HDL concentration in plasma.

✓ **ApoAII** is also present primarily in the HDL.

➤ It is a 77-a.a protein synthesized mostly in the liver.

➤ It inhibits lipoprotein lipase (LPL)

➤ serves as a cofactor for **LCAT** and

○ for the **cholesterol ester transfer protein (CETP)**.

❖ **Apolipoprotein B** exists in two common variants, **apoB100 & 48**

✓ ApoB100 is a large protein with a molecular mass of 513,000 kDa, comprising 4509 a.a.

✓ It is synthesized in the liver.

❖ **Apo B100** is present in VLDL, IDL, & **LDL**.

- ✓ It controls the metabolism of the LDL.
- ✓ Because there is only one molecule of **apoB100** per each LP particle, the measurement of the apoB in plasma reflects the sum of the VLDL, IDL, and LDL.
- ✓ **ApoB100** binds to the LDL (apoB/E) receptor.
- ✓ Mutation at it's a.a residue 3500 decreases its receptor binding and is the cause of a condition known as **familial defective apoB (FDB)**.

Cont...

❖ **ApoB48** is present in the chylomicrons.

- ✓ It is a truncated form of apoB100 synthesized from the same gene.
- ✓ An apoB100 mRNA translation is stopped by substituting the CAA codon that encodes glutamine with the codon UAA
- ✓ It is synthesized by the **enterocytes**.
- ✓ Note that apoB48 does not bind to the LDL receptor.
- ✓ Its plasma concentration reflects the sum of chylomicrons & chylomicron remnants.

❖ **Apolipoprotein E** is present in all lipoprotein classes.

- ✓ It has a molecular mass of 34,200 Da and comprises 299 a.a
- ✓ It binds to the LDL receptor with higher affinity than apoB100.
- ✓ It also binds to the LDL-receptor-related protein (LRP).
- ✓ ApoE stimulates the **LPL, the hepatic triglyceride lipase (HTGL), and the LCAT.**
- ✓ Its synthesis is controlled by three major alleles, $\epsilon 2$, $\epsilon 3$, and $\epsilon 4$, and it exists in three isoforms, E2, E3, and E4.

- ✓ The **E2 isoform** results from cysteine-for-arginine substitution at position 158 in E3 and has **lower affinity to receptors**.
- ✓ Thus in E2/E2 homozygotes, the uptake of the remnant particles is impaired and results in **familial dyslipidemia**
- ✓ In the HDL, **apoE contributes to cholesterol removal from cells**.
- ✓ In the brain, ApoE is synthesized by astrocytes and microglia:-
 - it affects growth and repair of CNS cells.

- ✓ Individuals with E4 phenotype were shown to be at an increased risk of the sporadic form of **Alzheimer's disease**.
- ✓ ApoE also has **anti-inflammatory and antioxidant** functions.
- ✓ ApoE phenotyping and genotyping is used in the diagnosis of familial dyslipidemia.

❖ ***Apolipoproteins C*** (apoCI, apoCII, and apoCIII)

- ✓ are enzyme activators and inhibitors, and they are extensively exchanged between different lipoprotein classes.

❖ *Apolipoproteins C-I* (apoC-I)

- ✓ It associated with Chylomicrons, HDL, VLDL, IDL
- ✓ may also activate LCAT; clustered with APOC2 and APOE genes on chromosome 19

❖ *Apolipoproteins C-II* (apoC-II)

- ✓ It associated with Chylomicrons, HDL, VLDL, IDL
- ✓ activates lipoprotein lipase; clustered with APOC1 and APOE genes on chromosome 19

❖ *Apolipoproteins C-III* (apoC-III)

- ✓ It associated with Chylomicrons, HDL, VLDL, IDL
- ✓ inhibits **lipoprotein lipase** and **hepatic lipase**,
- ✓ interferes with hepatic uptake and catabolism of apoB-containing LPs,
- ✓ appears to enhance the **catabolism of HDL** particles,
- ✓ enhances monocyte adhesion to vascular endothelial cells, activates inflammatory signaling pathways

❖ **Apolipoprotein (a), or apo(a),**

- ✓ possesses a protease domain and a number of repeating sequences of approximately 80–90 amino acids in length.
- ✓ Apo(a) is synthesized in the liver, and it binds to the LDL receptor.
- ✓ It is structurally related to plasminogen.
- ✓ Apo(a) is a component of lipoprotein (a), lp(a).
- ✓ Lp(a) is an LDL-like particle where apo(a) is covalently bound to apoB100.
- ✓ Lp(a) is modestly associated with *cardiovascular risk*.

LIPOPROTEIN RECEPTORS

❖ The LDL receptor

- ✓ is regulated by the intracellular **cholesterol concentration**
- ✓ Cellular uptake of lipoproteins is mediated by membrane receptors.
- ✓ The key lipoprotein receptor is the LDL receptor, also known as the **apoB/E receptor**.
- ✓ As its name indicates, it can bind either apoB100 or apoE.
- ✓ It was discovered by Joseph Goldstein and Michael Brown, who jointly received the Nobel Prize for this work in 1985.

❖ The LDL receptor

- ✓ Mature receptor protein contains 839 amino acids and spans the cell membrane.
- ✓ The receptor gene is located on **chromosome 19**, &
- ✓ its expression is regulated by the **intracellular free-cholesterol concentration**.
- ✓ Scavenger receptors are nonspecific and nonregulated
- ✓ Scavenger receptors are present on phagocytic cells such as macrophages.
- ✓ They are designated class A, class B, and CD36.

❖ The LDL receptor

- ✓ Scavenger receptors can bind many different molecules.
- ✓ They do not bind intact LDL but readily bind LDL that has been acetylated or oxidized.
- ✓ Class B receptor binds HDL particles in the liver.
- ✓ Importantly, scavenger receptors are not subject to feedback regulation, and therefore they may overload a cell with their ligand.

ENZYMES AND LIPID TRANSFER PROTEINS

- ✓ Two hydrolases, *lipoprotein lipase (LPL)* and *hepatic triglyceride lipase (HTGL)*, remove TAGs from LP particles.
 - **LPL** is bound to heparan sulfate proteoglycans on the surface of **vascular endothelial cells**, and
 - **HTGL** is associated with **liver** plasma membranes.
- ✓ **LCAT** associates with the HDL and esterifies cholesterol acquired by HDL from cells.

- ✓ Inside the cells, however, cholesterol is esterified by a different enzyme - **the acyl-CoA: acyl-cholesterol transferase (ACAT)**.
- ✓ There are two isoforms of ACAT:
 - **ACAT1** present in macrophages and
 - **ACAT2** present in the intestine and liver.
- ✓ **Cholesterol ester transfer protein (CETP)** facilitates the exchange of cholesteryl esters for TAGs between the HDL & other LPs.

PATHWAYS OF LIPOPROTEIN METABOLISM

✓ **Lipoproteins fulfill a dual function:**

I. Distribution of TAGs &

II. Cholesterol delivery to cells

✓ **TAGs** are transported toward the periphery for long-term storage in adipose tissue.

✓ **Cholesterol**, conversely, is moved in both directions (its transport from the peripheral tissues back to the liver is known as “reverse transport”).

PATHWAYS OF LIPOPROTEIN METABOLISM

- ✓ Cholesterol present in the lipoproteins forms an extracellular pool available to cells through the LDL receptor.
- ✓ Cholesterol delivered to the liver can be excreted in the bile.

Lipoprotein metabolism: The fuel distribution stage

- ✓ **In the fed state**, TAGs are delivered from the intestine to the periphery by **chylomicrons**;
- ✓ **Chylomicron remnants** form after TAGS are removed
- ✓ **After a fat-containing meal**, TAGs present in food are digested & absorbed in the intestine as **MAG & FFA**.
- ✓ Then the enterocyte-synthesized TAGs, together with
 - **phospholipids and cholesterol**, are assembled on the template of apoB48 into **chylomicron particles** by **MTTP**.

Cont....

- ✓ These are secreted into the lymph and reach enter the circulation at the left subclavian vein
- ✓ They also acquire **apoA, C, & E** from **HDL**
- ✓ Chylomicrons give plasma a milky appearance, and their half-life is less than 1 hour.
- ✓ In the peripheral tissues, TAGs are hydrolyzed by the **LPL**, & the FA enter cells.
- ✓ Chylomicrons depleted of TAGs become **chylomicron remnants**.

Cont....

- ✓ **TAGs** synthesized in the liver are transported to the periphery by the **VLDL**; this happens in **both fed & fasting states**
- ✓ **VLDL** are assembled in the liver on **apoB100** molecules in a process facilitated by **MTTP**.
- ✓ After being secreted into plasma, VLDL acquire **cholesteryl esters, apoC, and apoE** from the HDL.

Cont....

- ✓ Their **apoB100 and apoE** remain in conformations that do not allow binding to the LDL receptor.
- ✓ In the peripheral tissues, analogously to chylomicrons, VLDL TAGs are hydrolyzed by the **LPL**.
- ✓ VLDL depleted of triglycerides become **VLDL remnants** (also known as IDL).

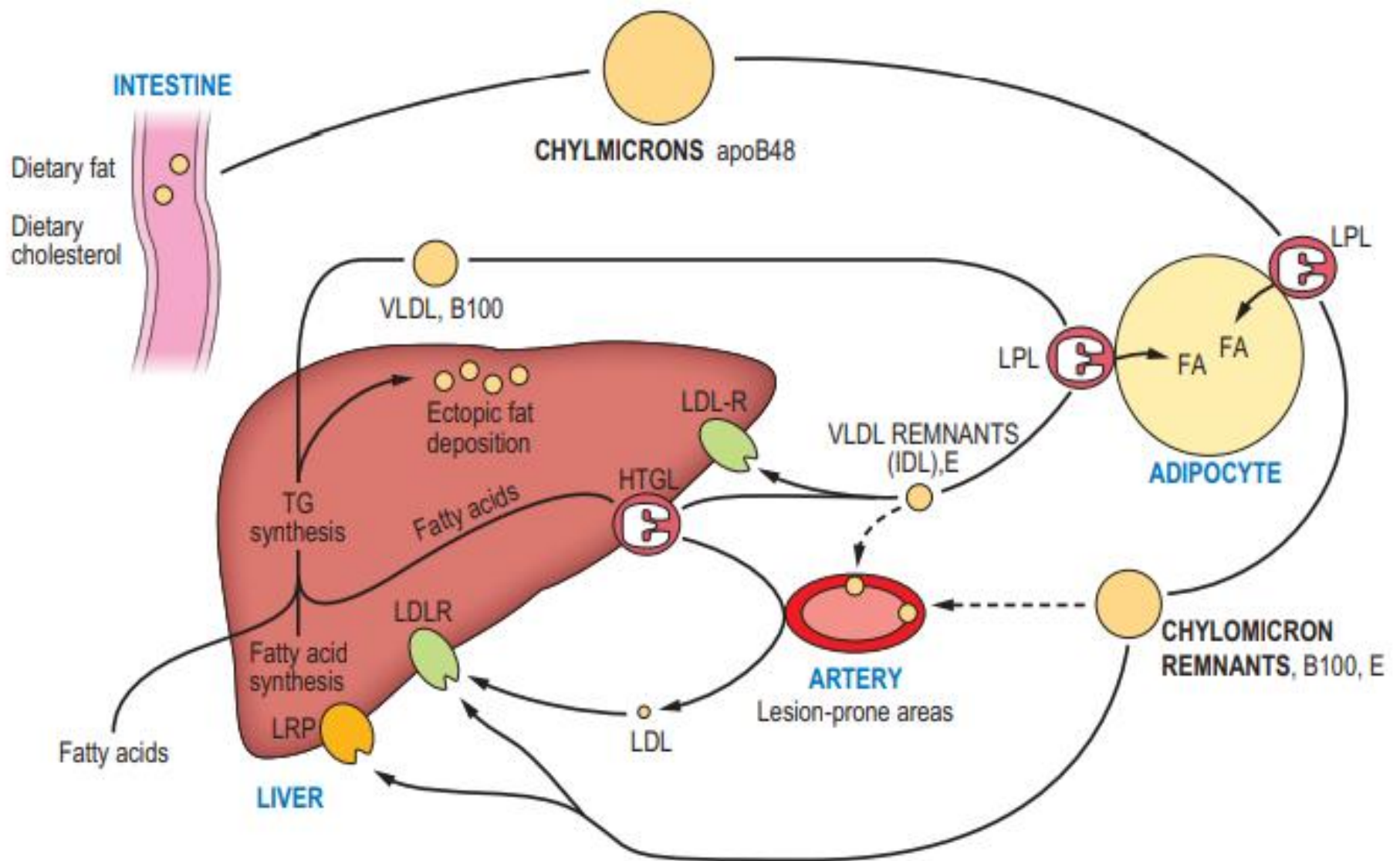


Figure: Lipoprotein metabolism: the fuel distribution stage.

Lipoprotein metabolism: The cholesterol delivery stage

- ✓ **Cholesterol** present in the **remnant particles** and in the **LDL** is transported to the liver. Chylomicron and VLDL remnants are smaller and denser than their precursors.
- ✓ The **remnant particles** are relatively cholesterol rich.
- ✓ In the chylomicron remnants, the change in particle size uncovers the apoE, so they can bind to LRP and to the LDL receptor.
- ✓ Similarly, in the VLDL remnants, apoE and apoB assume conformations that allow binding to the LDL receptor.

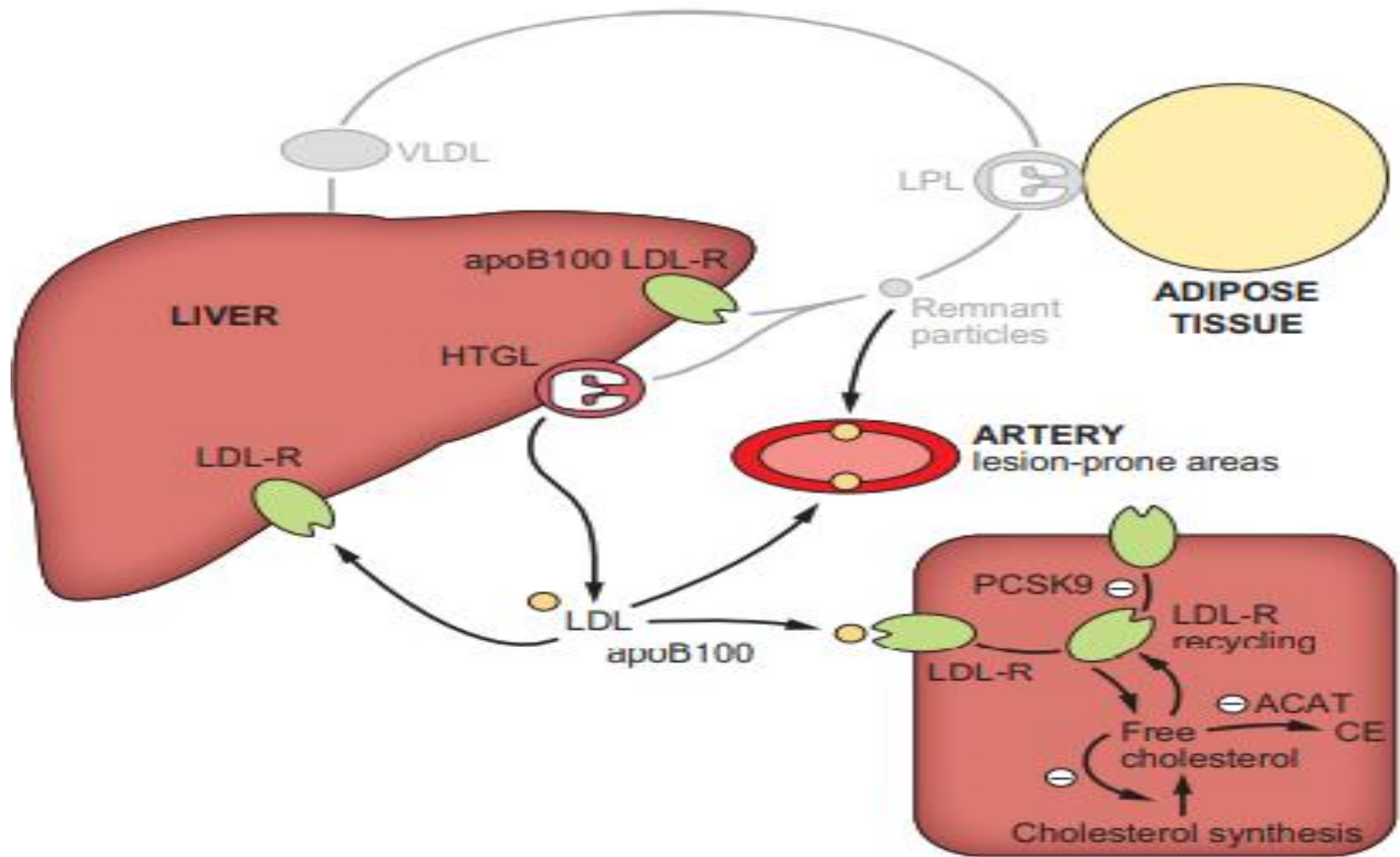
- ✓ **The VLDL remnants** are either taken up by the liver or depleted of remaining TAGs by the HTGL, and
 - Thus instead of being taken up, they transform into LDL.
- ✓ The LDL are released to plasma and eventually are internalized by cells after binding to the LDL receptor.
- ✓ About 80% of LDL is taken up by the liver; the rest is taken by peripheral tissues.

- ✓ After internalization, the LDL–receptor complex is digested by the lysosomal enzymes.
- ✓ The released free cholesterol is esterified within the cell, and the receptor protein **recycles back to the membrane**.
- ✓ The cholesterol distribution stage of lipoprotein metabolism.
- ✓ Plasma lipoprotein cholesterol forms an extracellular pool available to cells

- ✓ Most cells can synthesize cholesterol for their own needs.
- ✓ The concentration of free cholesterol in cell membranes regulates the way a cell acquires cholesterol.
- ✓ Free cholesterol exerts negative feedback on cellular cholesterol synthesis.
- ✓ This is mediated by transcription factors known as sterol regulatory element-binding proteins (SREBP)

- ✓ SREBPs regulate the transcription of genes coding for key enzymes in cholesterol synthesis –
 - the 3-hydroxy-3-methylglutaryl coenzyme A synthase &
 - HMG-CoA reductase –
 - but also of the gene coding for the LDL receptor.
- ✓ Cellular depletion of free cholesterol increases the SREBPs level, and consequently, both cholesterol synthesis and expression of the LDL receptor increase.

- ✓ Conversely, when a cell is replete in cholesterol, the SREBP pathway is inhibited, decreasing cholesterol synthesis and receptor expression.
- ✓ Thus the cholesterol contained in plasma lipoproteins is linked to its cellular content by a precisely regulated system.



Figure; Lipoprotein metabolism: cholesterol distribution stage

Reverse cholesterol transport

- ✓ HDL particles remove cholesterol from cells
- ✓ The HDL are assembled in **the liver and the intestine.**
- ✓ Their main apolipoproteins are **apoA1 and apoAII**, but they also contain **apoC and apoE.**
- ✓ HDL acquire cholesterol from peripheral cells & either transport it
 - ✓ by a direct route to the liver or
 - ✓ indirectly transfer it to TAG-rich particles, chylomicrons, VLDL, or LDL, where they follow the remnant/LDL route

Cont...

- ✓ The HDL are formed as **discoid**, lipid-poor particles (pre- β -HDL) that contain mainly apoA1.
- ✓ They are partly constructed from the excess phospholipids shed from the VLDL during their hydrolysis by LPL.
- ✓ They accept cholesterol from cells through the action of a membrane ATP-binding cassette transporter A1 (ABCA1)

Cont...

- ✓ Cholesteryl esters move into the interior of the particle, which enlarges and becomes **spherical** - it is now designated HDL-3.
- ✓ Aided by the **CETP**, it transfers some of its cholesteryl esters to chylomicrons, VLDL, & remnant particles in exchange for TAGs.
- ✓ Acquisition of TAGs enlarges the particle further - it now becomes HDL-2.

Cont...

- ✓ ABCA1 uses ATP as a source of energy & is rate-limiting for the efflux of free cholesterol to apoAI.
- ✓ Transfer of cholesterol from HDL to TAG-rich particles is the principal route of cholesterol transport in humans
- ✓ The free cholesterol acquired by the nascent HDL is esterified by **LCAT**.

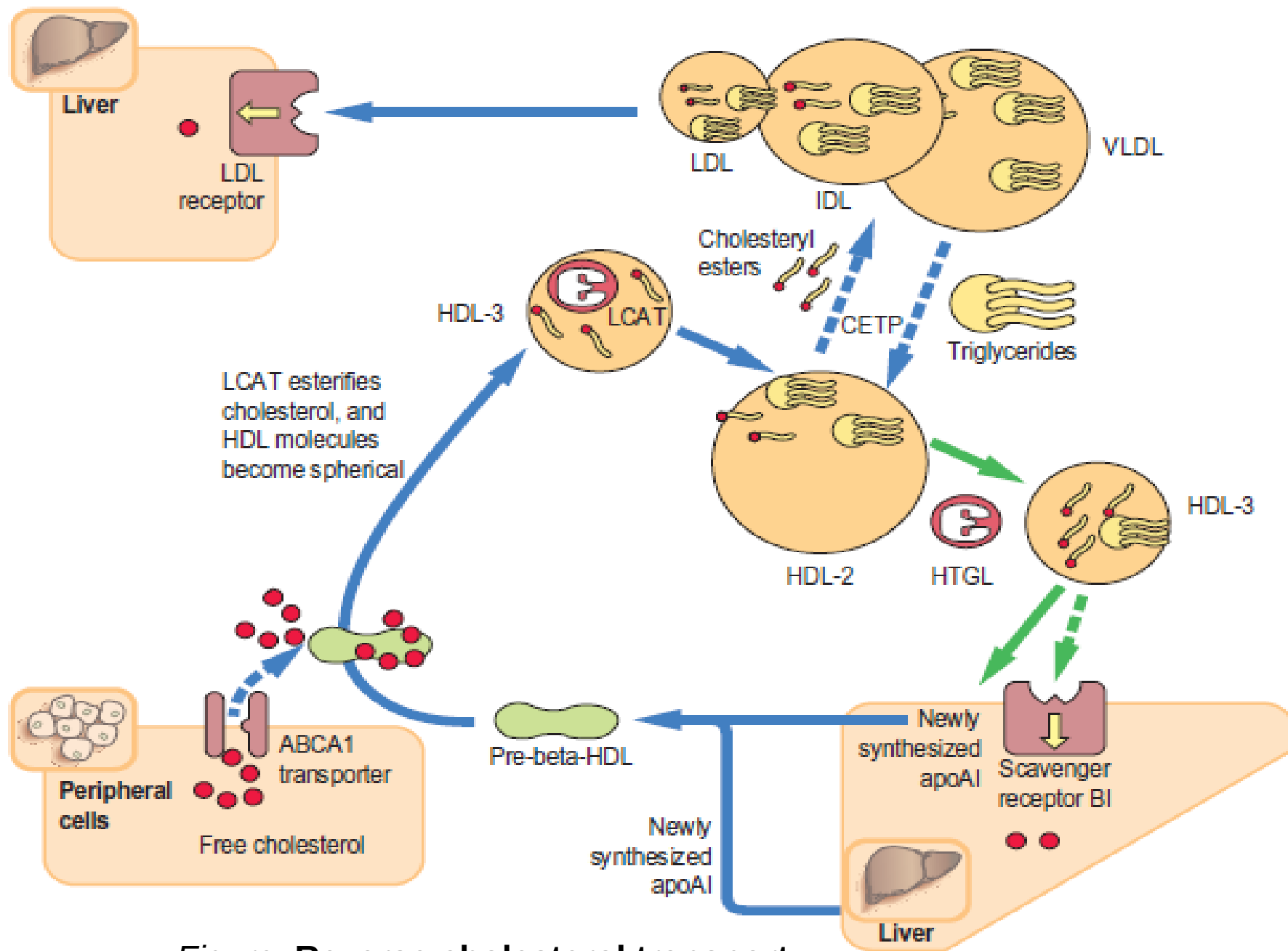


Figure: Reverse cholesterol transport.

Hyperlipidemia

abnormally elevated levels of any or all lipids and/or lipoproteins in the blood but Plasma cholesterol and triglyceride are clinically important because they are major modifiable risk factors for **cardiovascular disease**, whilst severe hypertriglyceridaemia also predisposes to **acute pancreatitis**.

Dyslipidaemia

- ❖ Defects in lipoprotein metabolism lead to disorders known as **dyslipidemias**, synonymously but less accurately called hyperlipidemias
- ❖ Their original, now outdated, classification into **types I to V** was based on the electrophoretic behavior of lipoproteins
- ❖ Another commonly used classification is phenotypic, which simply divides dyslipidemias into **hypercholesterolemia**, **hypertriglyceridemia**, and **mixed dyslipidemia**.

Table 33.4 Phenotypic classification of dyslipidemias

Dyslipidemia type (Fredrickson)	Increased electrophoretic fraction (lipoprotein type)	Increased cholesterol	Increased triglycerides
I	Chylomicrons	Yes	Yes
IIa	Beta (LDL)	Yes	No
IIb	Pre-beta and beta (VLDL and LDL)	Yes	Yes
III	"Broad beta" band (IDL)	Yes	Yes
IV	Pre-beta (VLDL)	No	Yes
V	Pre-beta (VLDL) plus chylomicrons	Yes	Yes

Type of Dyslipidemia

Disorder	Defect	Comments
Type 1a Hyperlipoproteinemia or Familial chylomicronemia syndrome(Familial LPL deficiency)	Mutations in lipoprotein lipase (LPL) gene	✓ Slow chylomicron clearance, reduced LDL and HDL levels; ✓ Characterized by very severe hypertriglyceridemia, episodes of abdominal pain, recurrent acute pancreatitis, eruptive xanthomas, hepatosplenomegaly; ✓ Treated by low fat/complex carbohydrate diet; no increased risk of coronary artery disease
Type 1b hyperlipoproteinemia (Familial apoC-II deficiency)	defect in APOC2 gene	✓ Loss of apoC-II function leads to reduced lipoprotein lipase activity; results in slow chylomicron clearance, reduced LDL and HDL levels; ✓ Characterized by very severe hypertriglyceridemia, episodes of abdominal pain, recurrent acute pancreatitis, eruptive xanthomas, hepatosplenomegaly; ✓ Treated by low fat/complex carbohydrate diet; no increased risk of coronary artery disease

Type of Dyslipidemia cont...

Disorder	Defect	Comments
Hyperlipoproteinemia type 1D	mutations in the GPIHBP1 gene	✓ GPIHBP1 encoded protein required for movement of LPL from cells of adipose tissue, heart, and skeletal muscle to surface of endothelial cells of vasculature in these tissues; ✓ Loss of LPL activity results in severe chylomicronemia and hypertriglyceridemia
Type 2a hyperlipidemia (Familial hypercholesterolemia,)	✓ LDL receptor defect mutations in the APOB gene ✓ Gain-of-function mutations in the PCSK9 gene	✓ Reduced LDL clearance leads to hypercholesterolemia (dramatic elevation in LDL cholesterol), resulting in early onset atherosclerosis and coronary artery disease

Disorder	Defect	Comments
<u>Familial ligand-defective apoB</u>	mutations in APOB gene in the LDL receptor-binding domain: two most common are Gln for Arg or Cys for Arg	✓ Significant reduction in affinity of apoB-100 for the LDL receptor; dramatic increase in LDL levels; no affect on HDL, VLDL or plasma TAG levels; ✓ Significant cause of hypercholesterolemia and premature coronary artery disease; ✓ often considered as a familial hypercholesterolemia
Type 3 hyperlipidemia ; remnant removal disease, broad beta disease, apolipoprotein E deficiency (Familial dysbetalipoproteine mia)	APOE gene mutations; patients only express the apoE2 isoform	✓ ApoE2 isoform interacts poorly with these receptors; results in severely reduced hepatic remnant clearance; ✓ causes tendon xanthomas, hypercholesterolemia and atherosclerosis in peripheral and coronary arteries due to elevated levels of chylomicrons and VLDL

Disorder	Defect	Comments
Type 4 hyperlipidemia (Familial hypertriglyceridemi a)	elevated production of VLDL associated with glucose intolerance	✓ Frequently associated with non-insulin dependent diabetes mellitus (type 2 diabetes) , obesity , chronic alcohol consumption or administration of progesterone and related hormones; elevated total serum cholesterol as a result of increased VLDL
Familial hyperlipoproteinemi a type 5	elevated chylomicrons and VLDL	results from mutations in the APOA5 gene; hypertriglyceridemia and hypercholesterolemia with decreased LDL and HDL
Primary (familial) hyperalphalipoprotei nemia 1: primary HALP1 (type 2b hyperlipidemia)	mutations in the CETP gene	increased level of HDL; a rare condition that is beneficial for health and longevity; a related disorder, HALP2, results from mutations in the apoC-III (APOC3) gene

Disorder	Defect	Comments
Familial LCAT deficiency (Norum disease) Fish-eye disease	complete (familial LCAT deficiency) or partial (Fish-eye disease) loss of LCAT	✓ Absence of LCAT leads to inability of HDL to take cholesterol from cells due to loss of esterification capability; ✓ Abnormal LDL (Lp-X) and VLDL; diffuse corneal opacities, swelling of optic nerve, accumulation of fat under the skin around the eyes (xanthelasma), target cell hemolytic anemia, and proteinuria with renal failure
Wolman disease and cholesteryl ester storage disease (CESD)	mutations in lysosomal acid lipase (LIPA) gene	✓ Lysosomal acid lipase is also called cholesteryl ester hydrolase; ✓ Wolman disease is severe form of LIPA deficiency (no enzyme activity), CESD mild form (some residual enzyme activity); Wolman disease associated with massive infiltration of organs with macrophages filled with triglycerides and cholesteryl esters, death occurs early in life; CESD is slowly progressing disease that primarily affects the liver

Type of Hypolipoproteinemias

I. Abetalipoproteinemia: ABL: mutations in the MTTP gene encodes microsomal triglyceride transfer protein;

- loss of function results; intestine and liver accumulate VLDL and chylomicrons, respectively;
- results in malabsorption of fat, steatorrhea, deficiency in fat soluble vitamins, retinitis pigmentosa, acanthocytosis (erythrocytes with a thorny appearance), and ataxic neuropathic disease

Type of Hypolipoproteinemias

- II. Familial combined hypolipidemia: mutations in ANGPTL3 gene encodes **angiopoietin-like protein 3**
- inhibits lipoprotein lipase and endothelial lipase activities;
 - therefore loss of activity results in enhanced fatty acid removal from circulating triglycerides;
 - ✓ associated with reduced levels of both apoB (VLDL/LDL) and apoA-I (HDL) containing lipoproteins;

Type of Hypolipoproteinemias

III. Tangier disease: mutations in the ABCA1 gene

- reduced HDL concentrations,
- no effect on chylomicron or VLDL production;
tendency to hypertriglycerideemia; some elevation
in VLDL; hypertrophic tonsils with orange
appearance

❖ What is **atherogenic triad**?

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Thank You